

EM Waves

What is a wave

We have a new differential forms which imply changing / propagating electromagnetic waves.

$$\begin{aligned}\vec{\nabla} \cdot \vec{E} &= 0 \\ \vec{\nabla} \times \vec{E} &= -\frac{\partial \vec{B}}{\partial t} \\ \vec{\nabla} \cdot \vec{B} &= 0 \\ \vec{\nabla} \times \vec{B} &= \mu_0 \vec{J} + \mu_0 \epsilon_0 \frac{\partial \vec{E}}{\partial t}\end{aligned}$$

Which leads to the result

$$\nabla^2 \vec{E} = \mu_0 \epsilon_0 \frac{\partial^2 \vec{E}}{\partial t^2}$$

We learned about waves propagating following the form

$$\frac{\partial^2 \vec{y}}{\partial x^2} = \frac{1}{v^2} \frac{\partial^2 y}{\partial t^2}$$

so the waves propagate at speed $\frac{1}{\sqrt{\mu_0 \epsilon_0}} = c$.

For a positive direction-propagating wave, the solution looks like $(x, t) = (x - vt)$ (recall that velocity is a vector).

The coupling that allows this to propagate is that changing electric fields cause changing magnetic fields. The Laplacian gives us some measure of the curvature of space in our field.

Properties of EM waves

Let

$$\begin{aligned}\vec{E}(x, y, z, t) &= \vec{E}_0 f(x-ct) \quad \text{where } \vec{E}_0 \text{ is a direction vector} \\ \end{aligned}$$

$$\vec{E}_0 = E_0 x \hat{x} + E_0 y \hat{y} + E_0 z \hat{z}$$

$$\begin{aligned}\vec{\nabla} \cdot \vec{E} &= 0 \quad \text{because we have } \rho = 0 \\ \implies \frac{\partial E_x}{\partial x} + \frac{\partial E_y}{\partial y} + \frac{\partial E_z}{\partial z} &= 0\end{aligned}$$

$$E_0 (x - ct)$$

$$\begin{aligned}\frac{d}{dx} E_0 \sin(x-ct) &= 0 \\ E_0 \cos(x-ct) &= 0\end{aligned}$$

$$E_{0,x} = 0 \quad \vec{\nabla} \times \vec{E} = -\frac{\partial \vec{B}}{\partial t}$$

$$\begin{aligned}\vec{\nabla} \times \vec{E} &= \frac{\partial}{\partial x} (E_0 \sin(x-ct)) \hat{z} = -\frac{\partial B}{\partial t}\end{aligned}$$

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\begin{align}
\frac{\partial B}{\partial t} &= \frac{\partial}{\partial x} (E_0 \sin(x-ct)(-\hat{z}) \backslash \\
\frac{\partial B}{\partial t} &= E_0 \cos(x-ct)(-\hat{z}) \backslash \\
\vec{B} &= \frac{E_0}{c} \sin(x-ct)(\hat{z}) \\
\end{align}

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$$Wvsntttntcnctcsntntntctns, nvntB_0 = \frac{E_0}{c} Bytnvsnttn, tntnntctntv, ttstctnBnnntctnE Wttvttyttnc$$

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\begin{align}
\vec{E} &= E_0 \hat{y} \sin(kx - \omega t) \backslash \\
k &= \text{wave number} \frac{2\pi}{\lambda} \backslash \\
\omega &= \text{angular frequency} = \frac{2\pi}{P} \text{osc} \\
\end{align}

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\begin{align}
&\text{this implies } \\
\frac{\omega}{k} &= c \\
\end{align}

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